

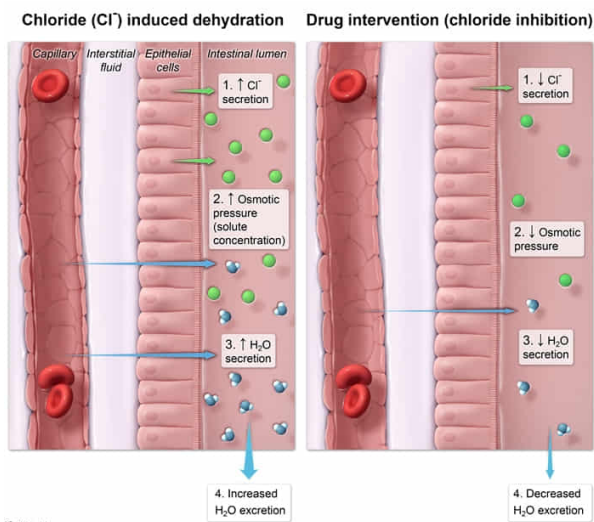
Dehydration caused by frequent loose stools is often observed in patients with certain diseases of the gastrointestinal (GI) tract. Given this, which of the following treatments would be most effective in treating this symptom?

- ☐ A. Use of drugs that increase smooth muscle contractions throughout the GI tract [6%]
- ☐ B. Use of drugs that decrease glucose absorption from the small intestine [10%]
- ☐ C. Use of drugs that increase activity of the parasympathetic nervous system [16%]
- ✔ ☒ D. Use of drugs that decrease the luminal secretion of chloride ions from intestinal epithelial cells [67%]

Correct

67% answered correctly

Explanation:



The **gastrointestinal (GI) tract** is composed of the esophagus, stomach, small intestine, and large intestine. Food broken down in the mouth travels down the esophagus and enters the stomach, which produces chyme, a semifluid mixture of partially digested food. Chyme flows into the **small intestine**, where **macromolecules** are further broken down and absorbed, along with water, by intestinal epithelial cells. The large intestine absorbs any remaining water and electrolytes.

In general, nutrients (ie, solutes) absorbed by intestinal epithelial cells are transported into surrounding interstitial fluid before diffusing into capillaries, *increasing* the osmotic pressure (solute concentration) in the interstitial fluid and, subsequently, in the blood plasma. Because water generally flows from low to high osmotic pressure via **osmosis**, *increased* osmotic pressure within interstitial fluid and blood plasma increases intestinal water absorption. In contrast, some epithelial cells *secrete* ions into the intestinal lumen, increasing osmotic pressure *within* the intestine to draw water in and lubricate luminal surfaces.

In this scenario, dehydration is caused by frequent loose stools in affected individuals. Loose (watery) stools may be due to *decreased* water absorption or *increased* water secretion in the intestinal tract. **Chloride (Cl⁻) secretion** from intestinal epithelial cells would *increase* the osmotic pressure of intestinal contents, drawing *more* water into the intestine and increasing the water content of stool. Therefore, drugs **inhibiting luminal secretion of Cl⁻** would likely *decrease* water secretion into the intestine and make the stools less watery.

(Choices A and C) **Parasympathetic signals** generally *enhance* **peristalsis** and motility of intestinal contents through the GI tract. Drugs that enhance parasympathetic activity and, consequently, peristalsis would *increase* the flow of chyme through the GI tract, *decreasing* the time available for nutrient and water absorption by intestinal cells and increasing (*not* decreasing) the excretion of watery stools.

(Choice B) A drug that *decreases* glucose absorption would **decrease osmotic pressure** of the interstitial fluid and capillary blood plasma surrounding the intestine. Decreased osmotic pressure caused by reduced glucose absorption would likely *decrease* intestinal water absorption and increase (*not* decrease) the excretion of water in stool.

Educational objective:

Most nutrient and water absorption occurs as food moves through the small intestine. The large intestine absorbs any remaining water and electrolytes prior to waste excretion.

Time spent:0 sec

Subject:
Biology

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System:
Digestion and Excretion